FEEL BEFORE YOU BUILD

Hands-on prototyping with actuators · 2 hours

AARHUS UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE

MULTIMODAL INTERACTION
PROTOTYPING WORKSHOP

GUSTAV SIMONSEN
TEACHING ASSISTANT

WHY WE ARE HERE

You can read a datasheet. You have never held one of these.

Most projects end up on the screen and the speaker

- Not because touch was wrong
- Because nobody knew a coin motor costs 12 kr

Today: decide it feels cheap, pick something else

- Better now than in week 46

HOW TODAY WORKS

A board each, five tasks, everything pre-wired

- Upload, change two numbers, upload again
- You wire nothing today

Actuators are on the benches around you

- Go and feel the ones your task does not use

Finish by building one signal

- Someone else has to read it

RUN SHEET

Times are offsets from the start. The only hard checkpoint is 0:25: everyone has task 01 running.

0:00	10 min	Two motors, same message. Which felt urgent?
0:10	15 min	Boards out, task 01 running for everyone
0:25	60 min	Tasks 02 to 05, at your own pace
1:25	10 min	Round-the-room: what surprised you
1:35	20 min	Blind test on task 05
1:55	5 min	Pack down

FIVE TASKS

FIVE TASKS, ONE BOARD EACH

Upload, change two numbers, upload again. That loop is the point.

01	**Make something spin**	**coin motor**
02	**Make something tap**	**solenoid**
03	**Make something warm**	**Peltier tile**
04	**Make it notice you**	**a wire and foil**
05	**Make it answer back**	**both together**

1 to 3 are output. 4 is input. 5 is both.

Nobody is expected to finish all five.

HOW THE TASK SKETCHES WORK

Every sketch runs the moment you upload it. Nothing to wire.

A CHANGE ME block at the top

- Two or three numbers. Edit, upload, feel the difference

A THINGS TO TRY list at the bottom

- Four experiments, roughly in order of difficulty
- The last one is usually the interesting one

github.com/gust1527/multimodal-actuator-workshop

THE ONE THAT SURPRISES PEOPLE

Touch input for the price of a wire. The thinking is where it costs you.

reading > baselineAverage() * 1.01

Threshold is relative, never absolute

- Readings drift with humidity and mains hum
- Fixed threshold works at 9am, fails at 1pm

Baseline only learns while untouched

- Or a long press teaches it that a finger is normal

INPUTS, AND PRIOR ART

INPUTS WORTH KNOWING ABOUT

One table with a sign. Wander over between tasks.

- Capacitive touch. A wire. That is task 04
- Heart rate (PPG). Useless once the hand moves
- Skin conductance (GSR). Relative change only
- Flex sensor. The cheap data glove
- Pressure (FSR). Calibrate every pad
- Your own phone. No soldering at all

TWO RIGS THAT ALREADY EXIST

Built here, by students at roughly your stage.

Moving screen, Arduino Uno

- Breathes when idle. Retreats when touched
- Easing and idle motion do the work. A dozen lines

Instrumented sock, ESP32

- Six force sensors under a foot, batched over Wi-Fi
- Calibrate per sensor. Batch your writes

BUILD ONE SIGNAL

THE BRIEF

30 minutes. Not long enough to be precious about it.

One actuator. Three messages

- Arrived. Something wrong. Finished

Vary two things only

- Rhythm of the pulses
- How strong they are

Blind test: they name all three

- Note which two got confused, and why

THE CONSTRAINT THAT MATTERS

The recipient cannot look at the device.

If it only works while someone watches a screen,

you built a visual interface with a motor glued to it.

GO AND TOUCH THINGS

github.com/gust1527/multimodal-actuator-workshop